

Unit: Unit 6: Systems of Linear Equations and Inequalities

Name: _____

Topic: Linear Inequalities with Two Variables

Date: _____

Challenge (Habanero)

Open Ended Questions (FRQ)

Q9. Graph the inequality $y < \frac{1}{2}x + 2$ on the coordinate plane. Identify the boundary line and shade the solution region.

Q10. A contractor is building a house with a rectangular foundation, where the length x and width y satisfy the inequality $x + 2y \leq 20$. If the length of the house must be at least 4 feet, write an inequality to represent the possible values of y . Graph the solution region on the coordinate plane.

Chef's Tips

- Check for correct graphing of inequalities
- Verify correct identification of boundary lines
- Ensure students provide clear and concise explanations for their answers